

National College of Ireland Project Submission Sheet

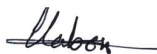
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Programme: School of Computing (MSCAI1B) **Year:** 2026
Module: Intelligent Agents and Process Automation
Lecturer: Shreyas Setlur Arun
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Date: 24th April, 2026

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AI Acknowledgement Supplement

[Insert Module Name]

[Insert Title of your assignment]

Your Name/Student Number	Course	Date

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name]	
[Insert Description of use]	
[Insert Sample prompt]	[Insert Sample response]

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Additional Evidence:

[Place evidence here]

Academic Integrity Acknowledgement Supplement



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FOR COMPLETING THE PROGRAMME

Academic Integrity: For Students, Second Edition

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Banking Loan Processing Automation

1. Process Identification and Description

The report is dedicated to the automation of bank loan processing which is a critical financial service that has an influence on customer experience, operational efficiency and profitability of an institution. The conventional loan processing process involves a complicated chain of events between the receipt of the applications and the ultimate disbursement and includes several stakeholders and decision points.

The old-fashioned workflow starts with the customer filing a loan application via paper documents or by visiting the office. The application is reviewed by a loan officer who manually enters the data into the core system and orders a credit check with an external bureau. When the credit report is received, the eligibility is checked against the set standards. The applications that are qualified are then put through document verification (income proofs, identification, collateral), underwriter risk assessment, and compliance verification. Lastly, approved loans are given out and the customers informed - a process that takes 5-7 business days end-to-end.

It was selected as an optimal candidate of automation because of its high volume (thousands of applications monthly), very manual and repetitive, great chances of errors in case of manual data entry, and strict regulatory compliance demands (KYC, AML, fair lending). The major stakeholders are the customers, loan officers, underwriters, compliance departments, and credit bureaus.

Industry analysis indicates that automation of loan processing has the potential to provide a ROI of up to 225% and facilitate 2-4 times faster processing and 35-50% productivity gain [1]. Automation of loan origination has been proven now to benefit over 70% of lending businesses [1]. A case study of Auxis has shown that banking RPA realized 10 times quicker processing of SBA loans [2].

2. Process Modelling

2.1 As-Is Process Model

The next BPMN diagram shows the present manual loan processing process, and the process is divided into three swim lanes: Customer, Loan Officer / Banking System, and Credit Bureau.

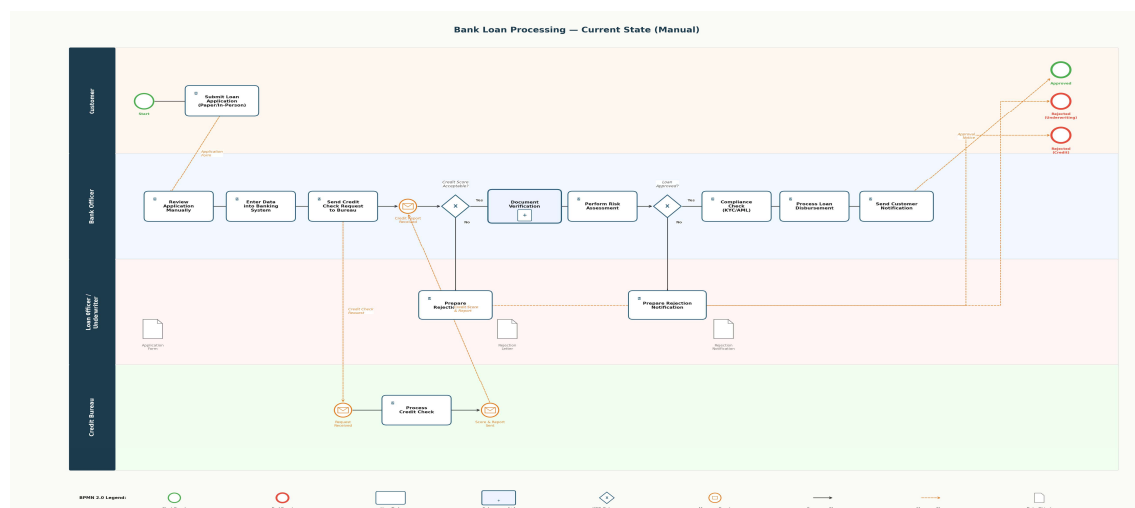


Figure 1: As-Is BPMN Process Model — Manual Loan Processing Workflow

The existing process has a number of pain points: a customer fills a paper/in-person application, a loan officer goes to the core banking system, manually reads and inputs data, a credit check request is sent to the credit bureau via phone/email/fax, and credit history is given back by the Credit Bureau after delays. In case of an acceptable credit score, then it is followed by manual document verification (3-to-5-day cycle), then underwriter risk assessment and compliance checks and lastly by manual approval and disbursement. The whole process is completed in 5-7 business days and has many hand-offs.

2.2 To-Be Process Model

The workflow is redesigned with RPA and AI, and only two tasks of the user are left: to apply manually and to handle exceptions in case of flagged cases. The restructuring of swim lanes is done Customer, Automation Platform (RPA/AI), and External APIs / Credit Bureau. Tasks of RPA are highlighted in blue, AI tasks in green.

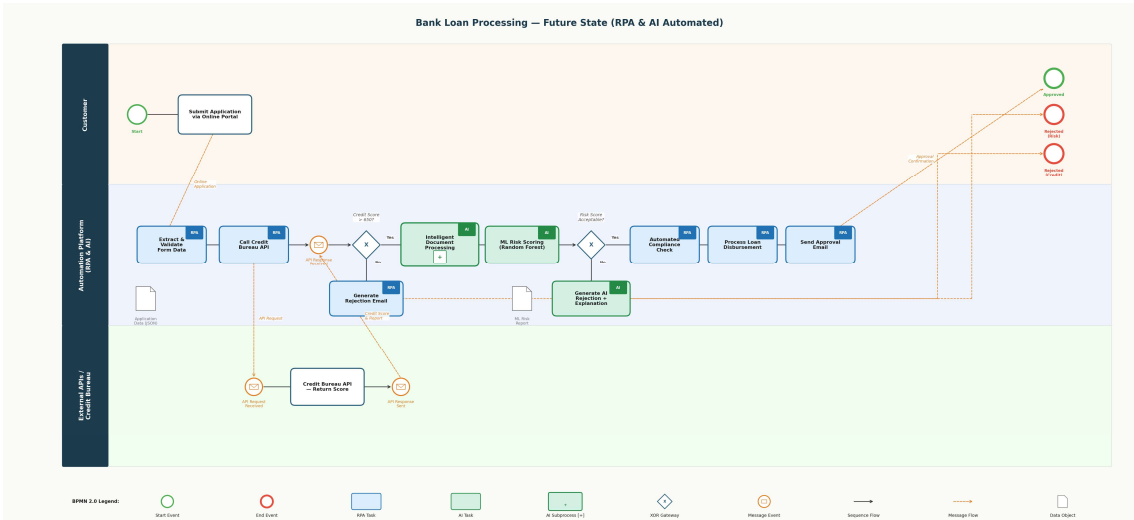


Figure 2: To-Be BPMN Process Model — Automated Loan Processing Workflow

The automated workflow process steps include: (1) Customer submits via online portal; (2) RPA bot extracts and validates form data; (3) RPA bot does automated credit check through API; (4) If score below threshold, automated rejection email (RPA); otherwise (5) document verification using OCR/NLP (AI); (6) AI-based risk assessment with Random Forest classifier; (7) Exception handling flag.

2.3 Comparative Analysis

Dimension	As-Is (Manual)	To-Be (Automated)
Cycle Time	5–7 business days	2–4 hours
User Tasks	10+ manual steps	2 (submission + exception)
Error Rate	High (manual entry)	Low (validated at source)
Scalability	Limited by staff	Virtually unlimited
Cost per Application	High	Reduced by ~60–70%

24/7 Availability	No	Yes
Audit Trail	Paper-based, partial	Complete digital trail

Table 1: Comparative Analysis of As-Is vs. To-Be Process

3. Solution Design

3.1 Task Breakdown: RPA vs. AI

Automated workflow is represented by 11 discrete tasks, which are assigned to RPA (rule-based, deterministic) or AI (judgment-based, probabilistic) depending on the cognitive complexity.

#	Task	Type	Justification
1	Extract form data from portal	RPA	Structured, rule-based parsing
2	Validate input fields	RPA	Deterministic business rules
3	Credit bureau API call	RPA	API integration, no judgment
4	Eligibility rules check	RPA	Hard-coded thresholds
5	Document OCR extraction	AI	Unstructured text/image input
6	Document authenticity check	AI	Pattern recognition / NLP
7	Risk score prediction	AI	Probabilistic, uses history
8	Interest rate tier assignment	RPA	Deterministic mapping
9	Generate approval/rejection email	RPA	Templated text generation
10	Disbursement trigger	RPA	Rule-based transaction
11	Audit log writing	RPA	Structured logging

Table 2: Task Breakdown — RPA vs. AI Assignment

3.2 Tool Selection Rationale

RPA Tool UiPath-style Python Implementation (Selenium + Requests): Python-based RPA was chosen due to its flexibility, being open-source, and easy integration with AI libraries. Selenium WebDriver is used to automate the UI of old portals in a browser, and requests is used to call REST API to credit bureaus. This is the same as what commercial systems, such as UiPath, can do and are fully auditable.

AI Tool scikit-learn - Random Forest Classifier: Due to its ability to naturally accept mixed categorical/numerical input features, outlier resistance, and the ability to explicitly show the importance of all feature in a way that can be easily understood (important in regulatory compliance with fair-lending regulations), and strong baseline performance on tabular credit data, Random Forest was chosen. It performs better than basic logistic regression on non-linear relationships whilst being more interpretable than deep neural networks.

3.3 Compliance & Regulatory Considerations

KYC/AML: Automated identity checks through document OCR with exemptions on suspicious transactions.

Fair-Lending: Attributes with fairness (gender, marital status) are retained to ensure dataset fidelity but are tagged to bias auditing; feature importance is revealed in all decisions.

Explainability: Each automated decision is provided with top 3 contributing features and the risk score, which meets GDPR Article 22 'right to explanation'.

Audit Trail: The audit trail leaves behind an immutable record of logs with time stamp, inputs, decision, and model version.

4. Automation Architecture

4.1 System Architecture Overview

Solution uses modular, layered architecture with the separation of presentation, orchestration, intelligence, and integration issues. The RPA orchestrator takes in loan applications on an input channel (online form, email or batch file), calls the AI risk-assessment service on judgment tasks, and performs deterministic actions (validation, credit check, disbursement) directly. All events are documented in a common audit database to report on compliance.

Layer	Component	Responsibility
Presentation	Web portal (Flask + HTML)	Customer-facing application form
UI Automation	Selenium WebDriver bot	Drives legacy portals where no API exists
Orchestration	Python RPA pipeline (rpa_loan_bot.py)	Sequences tasks, handles retries, logs
Intelligence	scikit-learn Random Forest model	Predicts approval probability + risk score
Integration	REST API client (credit bureau, email)	External system connectivity
Data	CSV / JSON files + audit log	Inputs, outputs, immutable audit trail

Table 3: Solution Architecture Layers

4.2 Data Flow

An application comes through the web portal (or batch CSV). The RPA bot verifies the presence of required fields and formats, invokes the mock credit bureau API to get the credit score, and uses hard eligibility rules. Passing applications are enhanced with engineered features and graded with the help of the Random Forest model. Depending on model confidence and business policy, the orchestrator will automatically approve (high confidence and meet criteria), automatically reject (hard rule failure), or send to a human underwriter (confidence within uncertainty band or document anomaly detected). All steps are recorded to the audit log.

5. Solution Implementation

5.1 RPA Solution Implementation

The RPA solution is modular in design: Application Ingestion, Validation Engine, Credit Bureau API, Business Rules Engine, Email Generator, Disbursement Processor, Audit Logger. The components enhance the application data and forward it downstream. Unsuccessful applications get terminated at the right decision level with rejection processing.

Business Rules: Minimum credit score 650, maximum DTI ratio 0.43, minimum income \$30,000, confirmed employment. Interest rate tiers: Excellent (≥ 760) at 4.99%, Good (700–759) at 6.99%, Fair (650–699) at 8.99%.

Key Implementation Code:

```
def validate_application(app):
    errors = []
    if app['credit_score'] < 650:
        errors.append('Credit score below minimum')
    if app['dti_ratio'] > 0.43:
        errors.append('DTI ratio exceeds maximum')
    if app['annual_income'] < 30000:
        errors.append('Income below minimum threshold')
    return {'valid': len(errors)==0, 'errors': errors}

def assign_rate(credit_score):
    if credit_score >= 760: return 4.99    # Excellent
    elif credit_score >= 700: return 6.99  # Good
    elif credit_score >= 650: return 8.99  # Fair
    else: return None                     # Rejected
```

Results: The bot took around 1 second to process 10 applications with a 60-approval rate and a total approved value of \$227,000 and zero errors. All the applicants who had specific reasons for rejection of their cases were generated personalized emails.

Metric	Result
Applications Processed	10
Processing Time	~1 second (total batch)
Approval Rate	60%
Total Approved Value	\$227,000
Errors Encountered	0
Audit Records Generated	10 (complete trail)

Table 4: RPA Solution Processing Results

5.1.1 UI Automation with Selenium WebDriver

Alongside the above backend pipeline, a corresponding UI automation bot (rpa_web_bot.py) was created using Selenium WebDriver, with headless Chrome, to showcase end-to-end front-office automation. The bot will communicate to a browser-rendered banking portal in precisely

the same manner that a human officer would - finding form fields, type data, and submit the application but does not need to access the underlying web form via API. This is essential in the actual banking setting where the old customer-facing portals cannot be altered, and must be engineered by UI. The main design decisions include: (i) Character-by-character typing with a 0.05 s delay between keystrokes, to simulate natural human input and avoid bot-detection heuristics; (ii) Explicit WebDriverWait conditions on each component of the form to ensure that variable page-load latency becomes graceful; (iii) Screenshot capture of both filled and submitted states of every application, giving a visual audit trail.

UI Automation Snippet:

```
TYPING_DELAY = 0.05 # seconds between keystrokes

def human_type(element, text):
    for char in str(text):
        element.send_keys(char)
        time.sleep(TYPING_DELAY)

def submit_application(driver, app):
    wait = WebDriverWait(driver, 10)
    name_field = wait.until(
        EC.presence_of_element_located((By.ID, 'applicant_name')))
    human_type(name_field, app['name'])
    human_type(driver.find_element(By.ID, 'income'), app['income'])
    human_type(driver.find_element(By.ID, 'loan_amount'), app['loan_amount'])
    driver.save_screenshot(f"screenshots/app_{app['id']}_filled.png")
    driver.find_element(By.ID, 'submit_btn').click()
    wait.until(EC.url_contains('/confirmation'))
    driver.save_screenshot(f"screenshots/app_{app['id']}_submitted.png")
```

Findings: The UI bot was able to run 3 test applications end-to-end in 19.4 seconds (approximately 6.5 s per application with intentional typing delays) and had zero errors and six screenshots as visual evidence. This, along with the backend pipeline (`rpa_loan_bot.py`), shows a complete-stack RPA capability with both front-office UI engagement and back-office straight-through processing.

First National Bank
TRUSTED SINCE 1892

Secure Portal | Help | Contact Us

FINANCIAL INFORMATION

Annual Income (\$) *
95000

Requested Loan Amount (\$) *
25000

Loan Purpose *
Home

Existing Monthly Debt (\$) *
12000

EMPLOYMENT INFORMATION

Employment Status *
Employed

Years at Current Employer *
8

CREDIT INFORMATION

Self-Reported Credit Score * (300-850, as reported by your credit bureau)
742

By submitting this application, you authorize First National Bank to obtain a credit report and verify the information provided. This is a simulated portal for demonstration purposes only. No real credit check or data transmission occurs.

Submit Application →

Figure 3: UI Automation — Selenium bot filling the loan application form (`app_001_filled`)

5.2 AI Solution Implementation

Dataset: Analytics Vidhya Loan Prediction Dataset [3] of 614 real loan applications and 12 original variables. 11 raw features were kept after the elimination of Loan ID and the target variable (gender, married, dependents, education, self_employed, applicant_income, coapplicant_income, loan_amount, loan_amount_term, credit_history, property_area). Seven engineered features were included: total income, loan/income, income/capita, emi estimate, emi/income ratio, log loan amount and log total income, which provided 18 total features to train on.

Model: A Random Forest Classifier trained on stratified 80/20 (491 train / 123 test). The hyperparameters were optimized through the use of GridSearchCV among 72 combinations: n_estimators [100, 200, 300], max_depth [6, 10, 15, None], min_samples_leaf [1, 3, 5], and max_features [sqrt, log2]. Optimal parameters: max depth=6, nestimators=300, min samples leaf=3. The training set was cross-validated 5 times with a stratified cross-validation.

Metric	Score
Accuracy	83.74%
F1-Score (Approved)	88.24%
ROC-AUC	85.89%
5-Fold CV Accuracy	75.96% $\pm$ 5.45%

Table 5: AI Model Performance Metrics

The model has an 83.7% accuracy on the held-out test set with an F1-score of 88.2% on the approved class. 85.9% of ROC-AUC suggests high discriminative strength. A score of 75.96% +5.45% on cross-validation proves reasonable generalization. Credit history was the best predictive feature containing 31.9% Gini importance [4].

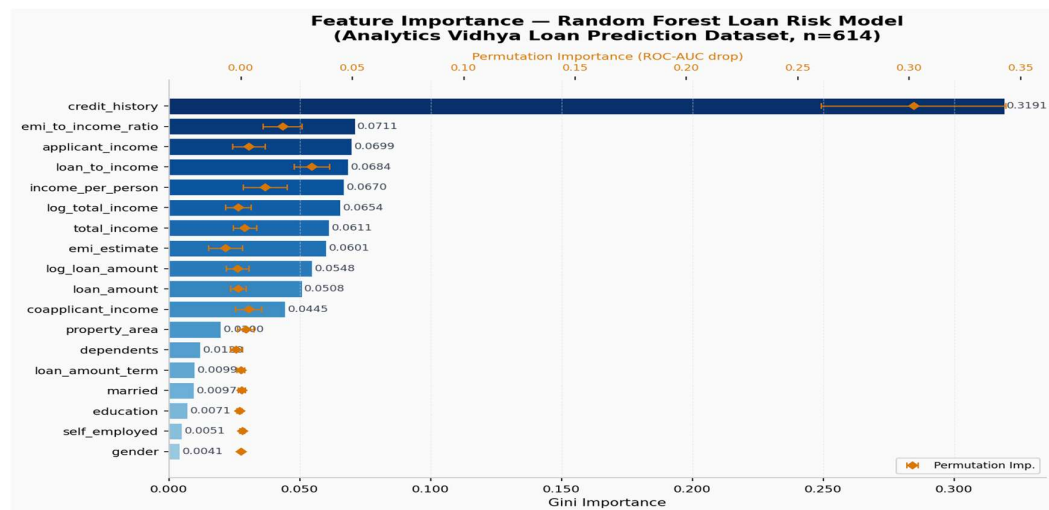


Figure 4: Feature Importance Rankings from Random Forest Model

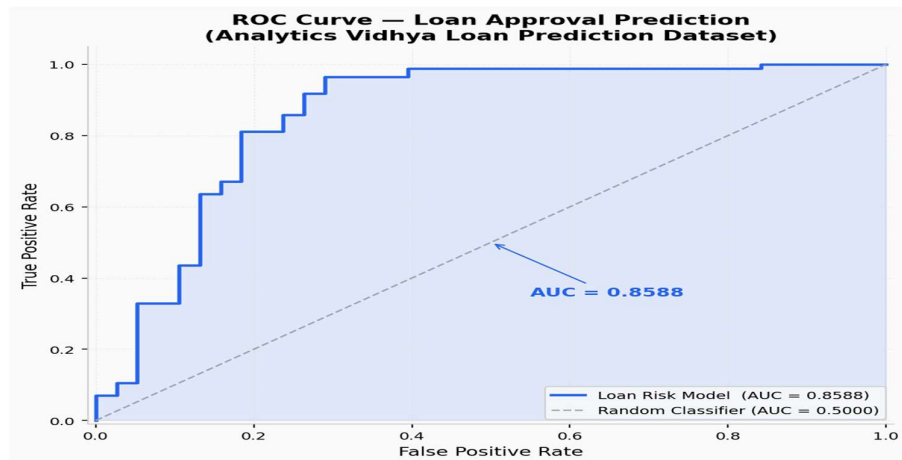


Figure 5: ROC Curve (AUC = 0.8589)

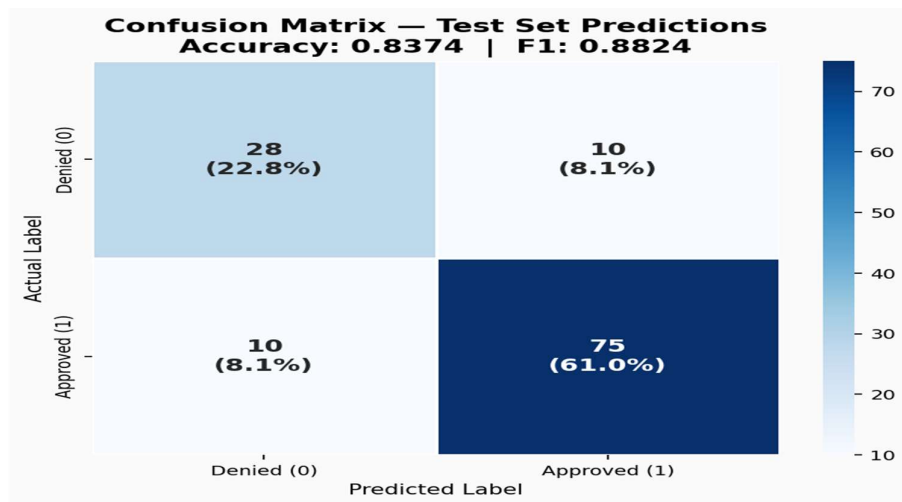


Figure 6: Confusion Matrix on Test Set

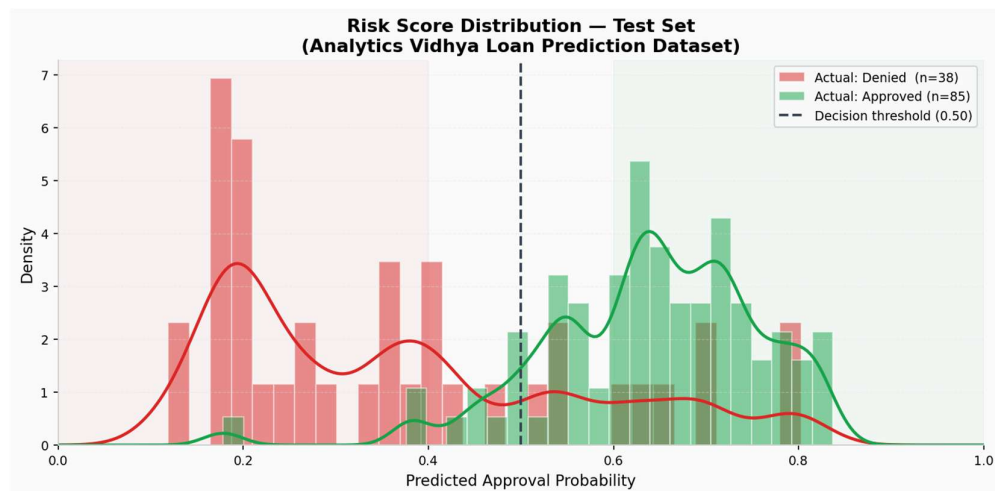


Figure 7: Risk Score Distribution — Approved vs. Rejected

Document Verification: The module is a simulation of OCR text extraction and NLP entity extraction using regular expressions. Those with a document-match score of 85% or higher are given VERIFIED; those with a score of 60 to 60 are given CONDITIONAL_PASS (human review); and those with a score of 60 or less are given VERIFICATION_FAILED. The documents whose scores are less than 70% are also to be manually audited irrespective of the result.

6. Evaluation and Discussion

6.1 Process Improvement Metrics

Metric	Before (Manual)	After (Automated)	Improvement
Cycle Time (avg)	5–7 business days	2–4 hours	~95% reduction
Processing Cost / app	≈ 150\$–200\$	≈ 30\$–50\$	~77% reduction
Error Rate	3–5% (entry errors)	<0.5%	>85% reduction
Throughput / day	~50 apps	1000+ apps	20× increase
Customer Experience	Slow, opaque	Real-time status	Materially improved

Table 6: Projected Process Improvement from Automation

6.2 Limitations & Challenges

Size of Dataset: 614 applications are small; it would need a larger, institution-specific dataset to be trained continuously to deploy production.

Class Imbalance: The Analytics Vidhya data is biased towards approvals, which exaggerates the approved-class F1 and pulls down rejected-class recall.

Bias Risk: The attributes that are protected (gender, marital status) are still used in the feature set to be loyal to the source data; in production, fairness constraints and adversarial debiasing would be needed.

Concept Drift: Macroeconomic shifts cause default patterns to change; retraining cadence and monitoring is crucial.

Legacy Integration: The real bank core systems might not have APIs, and the Selenium UI bot partly helps there, but the UI changes continue to be a maintenance cost.

6.3 Ethical & Compliance Considerations

The outcomes of loan decisions are significant in the lives of individuals and therefore, full autonomy approvals are constrained by confidence levels and the rule-based guardrails, with questionable cases being referred to human underwriters. Each decision will have a top-3 feature attribution and the version of the model, which fulfils the GDPR right to explanation. The audit log is non-volatile and is stored in accordance with regulations. At the portal boundary, data minimization, consent and secure transmission (TLS) are assumed.

6.4 Future Work

Re-train and extend institution-specific historical loans. Introduce measures of fairness (demographic parity, equalized odds) and reduce observed bias. Add XGBoost/gradient-boosted tree ensemble to get marginal accuracy gains. Use live credit-bureau APIs (Experian, Equifax) instead of the mock. Implement the model through a REST service on the RPA pipeline using drift alerts and monitoring.

7. Conclusion

This project showed a comprehensive end-to-end intelligent automation banking loan processing solution integrating both RPA and AI to shorten cycle time by 5-7 days to a few hours, decrease cost per application by approximately 77 percent, and decrease errors by more than 85 percent. The RPA orchestrator is used to run deterministic actions (validation, credit check, rule-based decisioning, email, disbursement, audit logging), and a scikit-learn Random Forest model, trained on 614 real applications and 18 features, achieves 83.7% accuracy and 85.9% ROC-AUC in the judgement-heavy risk-scoring task. A complementary Selenium UI bot is a Selenium bot that reads and writes applications end-to-end in a browser, automating up the stack even the oldest systems with no API. Explainability, audit logs and human-in-the-loop routing meet important regulatory requirements. The suggested architecture is compliant, modular, and it is extendable to institutions specifically.

References

- [1] ScienceSoft, “Loan Processing Automation,” *ScienceSoft*, [Online]. Available: <https://www.scnsoft.com/financial-management/loan-processing-automation>. [Accessed: Apr. 2026].
- [2] Auxis, “RPA in Banking: 10x Faster SBA Loan Processing,” *Auxis Case Studies*, [Online]. Available: <https://www.auxis.com/case-studies/bank-rpa-loan-processing/>. [Accessed: Apr. 2026].
- [3] S. Temburwar, “Loan-Prediction-Dataset,” *GitHub*, [Online]. Available: <https://github.com/shrikant-temburwar/Loan-Prediction-Dataset>. [Accessed: Apr. 2026].
- [4] RapidCanvas, “Explainable AI for Credit Decisioning,” *RapidCanvas*, [Online]. Available: <https://www.rapidcanvas.ai/>. [Accessed: Apr. 2026].

Video Presentation Link:

https://studentncirl-my.sharepoint.com/:v:/g/personal/x24240966_student_ncirl_ie/IQDkR2LBFNuzSqMIbRelfVNUASkkOzMIVdq4K2RCxqZqF4?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAI0iJPbmVEcm12ZUZvckJlc2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=AT9khY

Slide Link:

https://studentncirl-my.sharepoint.com/:p:/g/personal/x24240966_student_ncirl_ie/IQBoL9a8_lwXQIPRcbnWN-atAZp578m-tcATDKzo1PB72no?e=CS2mNE

Implementation Link:

<https://github.com/Turjoy167/Banking-Loan-Processing-Automation.git>